

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-76059AB0F5

Country: Tanzania

Submission Date: 2025-09-22

Product: Digovia (digoxin) | ATC: C09AA02 | Strength: 5 mg

Applicant: Prime Therapeutics | Manufacturer: PanRegional Pharma Holdings

Policy Label: standard\_review | Risk Score: 0.5

AMR Stewardship: AWaRe=watch | Unmet Need=not\_applicable | GLASS Trend=not\_applicable | Watch  
Similarity=not\_applicable

SECTION CONTENT

[1] m1\_application\_admin - Application Form and Administrative Information

Labels: presence=present; length=length\_ok; correctness=correct

This dossier constitutes a formal application for Marketing Authorization of Digovia (digoxin) submitted on 2025-09-22 to the national regulatory authority of Tanzania. The applicant, Prime Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 5 mg, classified under ATC code C09AA02. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. All referenced documents are available in the dossier annex and traceable by section identifier. The applicant confirms alignment with applicable technical guidance and regional submission standards. A comprehensive table of contents and glossary of abbreviations are provided to facilitate re

[1] m1\_manufacturer\_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length\_ok; correctness=correct

The primary commercial manufacturing site for the finished pharmaceutical product is PanRegional Pharma Holdings, located in Tanzania. The facility was subject to a comprehensive GMP inspection by a recognized stringent regulatory authority. The latest inspection concluded on 2024-09-13, resulting in a formal compliance status of 'compliant'. The active GMP certificate number is GMP-196214, which remains valid until 2026-12-28. Site master files, a summary of recent inspection observations, and evidence of completed Corrective and Preventive Actions (CAPA) are included in the annex. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Site quality-system commitments were compared against the proposed commercial manufacturing activities. The GMP evidence package includes certificate status, inspection scope, and manufacturing-site accountability records. The applicant confirms alignment with applicable technical guidance and regional submission standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Inspection history, CAPA closure records,

Site Inspection History Summary:

| Inspection Date | Authority | Scope | Outcome |
|-----------------|-----------|-------|---------|
|-----------------|-----------|-------|---------|

|            |               |             |           |
|------------|---------------|-------------|-----------|
| 2024-09-13 | National Auth | FPP/General | compliant |
| 2022-05-12 | SRA Joint     | API/Sterile | compliant |

#### [1] m1\_product\_information - Product Information and Naming

Labels: presence=present; length=length\_ok; correctness=correct

The proposed proprietary name for this medicinal product is Digovia, containing the active pharmaceutical ingredient digoxin. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. WHO AWaRe category: Watch. Similarity to existing Watch comparator not\_applicable is assessed as not\_applicable. Stewardship materials describe controlled use to limit cross-resistance pressure. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The container closure system consists of PVC/PVDC-Aluminum blis

#### [2] m2\_quality\_overall\_summary - Quality Overall Summary

Labels: presence=present; length=length\_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for digoxin. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1)

guidelines, demonstrating acceptable accuracy, precision, an

## [2] m2\_clinical\_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length\_ok; correctness=correct

The Clinical Overview synthesizes the efficacy and safety data supporting the use of Digovia for the treatment of bacterial skin infection. The pivotal clinical development program comprised 4 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is 'endpoint\_met'. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. GLASS-aligned surveillance trend is not\_applicable and cross-resistance risk relative to not\_applicable is assessed as not\_applicable. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication. Clinical interpretation focuses on whether efficacy, safety, and benefit-risk conclusions are supported by the submitted evidence. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. No new safety signals were identified during the extended open-label follow-up period. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The target population demographics are broadly representative of the intended commercial patient cohort. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. The risk management plan includes

## [3] m3\_api\_quality - API Quality and Control Strategy

Labels: presence=present; length=length\_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. All referenced documents are available in the dossier annex and traceable by section identifier. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Method validation records and quality controls were reviewed for internal consistency. Critical material attributes and control strategy elements were reviewed for consistency across the

quality dossier. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The API package links specification limits to validated analytical methods and impurity-control decisions. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Batch analysis, validation, and impurity management records support the proposed API quality framework. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The applicant confirms alignment with applicable technical guidance and regional submission standards. Pharmacokinetic

### [3] m3\_fpp\_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length\_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. The applicant confirms alignment with applicable technical guidance and regional submission standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. All referenced documents are available in the dossier annex and traceable by section identifier.

Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was

### [3] m3\_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length\_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. Pharmacokinetic

parameters were derived using non-compartmental analysis from the concentration-time profiles. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. All referenced documents are available in the dossier annex and traceable by section iden

#### [4] m4\_nonclinical\_summary - Nonclinical Study Summary

Labels: presence=present; length=length\_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accurac

#### [5] m5\_trial\_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length\_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (bacterial skin infection). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. GLASS-aligned surveillance trend is not\_applicable, and cross-resistance risk relative to not\_applicable is assessed as not\_applicable. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards

Pivotal Clinical Trial Inventory:

| Protocol ID | Phase | Enrollment | Status    |
|-------------|-------|------------|-----------|
| PROT-001    | III   | 628        | completed |
| PROT-002    | III   | 271        | completed |
| PROT-003    | III   | 693        | completed |
| PROT-004    | III   | 651        | completed |

#### [5] m5\_pivotal\_trial\_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length\_ok; correctness=correct

Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of 'endpoint\_met', demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. GLASS-aligned surveillance trend is not\_applicable, and cross-resistance risk relative to not\_applicable is assessed as not\_applicable. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. Method validation records and quality controls were reviewed for internal consistency. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. All refe

#### [5] m5\_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length\_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ( $f_2 > 50$ ). The bioanalytical methods used for plasma quantification were fully validated. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal

consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Wher